

Muhammad Rafi Syafrinaldi

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Executive Summary

Final-year Master's student in Computer Science with 3+ years of hands-on experience and 1+ year of professional experience in data science and machine learning. Led MLOps on AWS/GCP development of production-ready Generative AI systems including AI News, customer service chatbots, lead enrichment, and fraud detection. Specialized in NLP, LLM, RAG, and tabular data ML including predictive analytics, sentiment analysis, and time series modeling achieving 85-94% accuracy.

Skill

Programming Languages (Python, R, C++, SQL, JavaScript, Java, CSS, HTML)

Tools & Libraries (TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, LangChain, GPT, NLTK, FAISS, CodeBERT, XGBoost, Random Forest, Gradient Boosting, SVM, Naive Bayes, Airflow, Kafka, dbt, PostgreSQL, MongoDB, Redis)

Clouds & Infrastructure (AWS, GCP, Docker, FastAPI, REST APIs, n8n, Power BI, Tableau, Apache Superset)

Experience

CRIF Lembaga Informasi Keuangan (CLIK)

Jakarta, Indonesia

Analytics Consultant (Data Scientist)

Dec 2025 – Feb 2026

- Conducted market research and analysis to support CLIK's data-driven business strategies.
- Developed proof-of-concept (POC) data science solutions for business intelligence and analytics needs.

Caprae Capital Partners

Glendale, CA

Full-stack AI Engineer

Mar 2025 – Jan 2026

- Led and mentored **10+ team members** consisting of other **interns**, conducting **code reviews**, **technical workshop sessions**, and **project planning** to ensure high-quality delivery of Generative AI systems.
- Designed and implemented **AI News application** using **DeepSeek API** and **TF-IDF** for **news summarization**, **keyword tagging**, and **relevancy analysis** of scraped news from **RapidAPI**, creating automated content curation system with **real-time updates**.
- Developed **customer service ChatBot** with **multi-turn conversation capabilities** using **LangChain**, **FAISS**, and **Redis**, implementing **conversation memory**, **context retention**, and **conversation log analysis** to monitor **output quality** and **safety issues**, achieving **85% accuracy** in intent recognition and reducing hallucination rates by **40%** through **prompt optimization framework**.
- Built **AI-powered Lead Enrichment system** using **Python**, **Pandas**, **NumPy**, and **DeepSeek API** to classify **business types**, assess **relevancy**, and generate **lead scoring** for prospect prioritization, improving sales team efficiency by **45%**.
- Developed **Task Notifier Chatbot** using **n8n** automation platform with **Gmail**, **Slack**, and **Google Sheets** integration for workflow orchestration, implementing **task tracking** and **automated notifications**, cutting down production time by **50%** through streamlined communication.
- Built **Company Insights Dashboard Generator** using, **Pandas**, **NumPy**, **DeepSeek API**, **Playwright**, and **Selenium** to analyze **connected people**, **reviews**, and **growth trends**, implementing **AI tagging** and **competitor analysis** providing actionable business intelligence for strategic decision-making. Achieved **90% data processing accuracy** and reduced manual reporting time by **60%**.
- Deployed **production-ready conversational AI systems** with **API endpoints** using **FastAPI** and **Docker** on **AWS Lightsail** and **Amplify**, implementing **rate limiting**, **authentication**, and **monitoring** to serve **3,000+ users**.

Universitas Gadjah Mada (UGM)

Yogyakarta, Indonesia

Research Assistant

Feb 2025 – Aug 2025

- Conducted research on **Knowledge Management for COVID-19 mitigation** using **data analytics** and **statistical modeling**, processing **10,000+ data points** from **government databases**, **healthcare records**, and **social media** to improve data accuracy by **15%**.
- Developed **ETL/ELT data processing pipelines** using **Airflow**, **Kafka**, and **dbt** with **PostgreSQL**, **Python**, and **NumPy**, reducing processing time by **40%** and achieving **88% model accuracy** for knowledge classification.
- Built **dashboards using Power BI** with **feature engineering** and **logistic regression** models to track **COVID-19 knowledge gaps** and **mitigation effectiveness** in real-time.
- Performed **statistical analysis** using **Python** with **hypothesis testing** and **topic sentiment analysis**.

Universitas Islam Riau

Pekanbaru, Indonesia

Data Scientist

Feb 2023 – Jul 2023

- Designed automated **data processing pipeline** for **sentiment analysis** using **Python**, **Pandas**, **Scikit-learn**, **NLTK**, and **PostgreSQL** with **TF-IDF vectorization** to process large volumes of email data daily, creating an efficient **sentiment**

analysis system.

- Implemented comprehensive **model evaluation framework** using Naive Bayes, Logistic Regression, and SVM with **cross-validation (5-fold)** and **performance monitoring** to ensure robust **sentiment analysis model performance** and reliability.

Projects

Credit Risk Analysis and Loan Approval Prediction	2025
• Built comprehensive predictive analytics system using XGBoost and ensemble methods, implementing data preprocessing, feature engineering, and model training with 94% accuracy, demonstrating optimization and statistical modeling for financial risk assessment.	
Urban Heat Island (UHI) Predictor	2025
• Developed web application to predict urban heat island hotspots using Linear Regression, Random Forest, and Gradient Boosting with React frontend and FastAPI backend, implementing geospatial feature engineering, GridSearchCV hyperparameter tuning, and cross-validation (5-fold), achieving RMSE 0.8°C and R² 0.85.	
Fraud Detection RAG Chatbot	2025
• Designed conversational AI for anomaly detection using LLM-based tools (GPT-4) and LangChain with retrieval-augmented generation (RAG), implementing prompt engineering and context management achieving 90% accuracy, with API development using FastAPI.	
Wild Fire Detection System	2025
• Built computer vision system for image detection and tracking using neural networks (ResNet-50, YOLO v8) with TensorFlow, PyTorch, and OpenCV, achieving 89% accuracy through model training, data preparation, and inference optimization.	

Education

Master of Computer Science (Fast Track) , <i>Universitas Gadjah Mada – Yogyakarta, Indonesia</i>	Aug 2024 – July 2026 (expected)
• CGPA: 4.00/4.00 • Coursework: ML pipeline design, AI model deployment, ETL/ELT pipeline design, and scalable data processing.	
Bachelor of Computer Science (Cum Laude) , <i>Universitas Gadjah Mada – Yogyakarta, Indonesia</i>	Jul 2021 – Nov 2024
• GPA: 3.71/4.00 • Coursework: AI/ML, Deep Learning, Computer Vision, NLP, Data Mining, and Research Methods. • Thesis: "CodeBERT with Transfer Learning for Cross-Site Scripting Vulnerability Detection"	
Winter Abroad Program , <i>Yonsei University – Seoul, South Korea</i>	Dec 2023 – Jan 2024
• GPA: 3.77/4.00	

Certifications

• Associate AI Engineer , <i>Datacamp</i>	Oct 2025
• Data Science Bootcamp , <i>Dibimbing.id</i>	Dec 2022
• Deep Learning Specialization (TensorFlow and Keras) , <i>IBM</i>	Apr 2025
• Google Professional Cybersecurity Certificate , <i>Google</i>	Nov 2023

Achievements

Master DIKE Scholarship , <i>Department of Computer Science and Electronics FMIPA UGM</i>	Aug 2024
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Organizational Experience

Student Executive Board KM FMIPA UGM , <i>Universitas Gadjah Mada – Yogyakarta, Indonesia</i>	Jul 2021 – Nov 2023
• Head of Design Department - Managed team creating engaging social media content for BEMKM FMIPA UGM. • Head of Creative, Grand Launching Event - Directed team to design and produce materials for stakeholders.	

Languages

English: Professional Working Proficiency (IELTS 7.0, DET 135)	<i>Certification</i>
Indonesian: Native	Malay: Professional Working Proficiency